Math 162
Problem set 9 solutions

Justin Campbell

1. (a) Fix $\epsilon > 0$. Since $f_n \to f$ uniformly, there exists $N \in \mathbb{N}$ such that

$$|f_N(x) - f(x)| < \frac{\epsilon}{4(b-a)}$$

(one can also require the same inequality for any $n > N$, but we won't need this). For any partition P of $[a,b]$, let us write

$$M_i := \sup\{f(x) \mid x_{i-1} \leq x \leq x_i\} \quad \text{and} \quad M_i' := \sup\{f_N(x) \mid x_{i-1} \leq x \leq x_i\}$$

for any $1 \leq i \leq m$, where P has length m. Then we have

$$|M_i' - M_i| \leq \frac{\epsilon}{4(b-a)} < \frac{\epsilon}{3(b-a)}$$

and hence

$$|U(f_N, P) - U(f, P)| = \left|\sum_{i=1}^{n}(M_i' - M_i)\Delta x_i\right| \leq \sum_{i=1}^{n}|M_i' - M_i|\Delta x_i < \frac{\epsilon}{3(b-a)}\sum_{i=1}^{n}\Delta x_i = \frac{\epsilon}{3},$$

and similarly

$$|L(f_N, P) - L(f, P)| < \frac{\epsilon}{3}.$$

Using the integrability of f_N, we can choose P so that

$$U(f_N, P) - L(f_N, P) < \frac{\epsilon}{3},$$

from which we conclude that

$$\begin{aligned} U(f,P) - L(f,P) &= U(f,P) - U(f_N,P) + U(f_N,P) - L(f_N,P) + L(f_N,P) - L(f,P) \\ &\leq |U(f,P) - U(f_N,P)| + U(f_N,P) - L(f_N,P) + |L(f_N,P) - L(f,P)| \\ &< \frac{\epsilon}{3} + \frac{\epsilon}{3} + \frac{\epsilon}{3} = \epsilon. \end{aligned}$$

Thus f is integrable by Darboux's criterion.

(b) The easiest way is construct an example where f is unbounded. For example, let $f : [0,1] \to \mathbb{R}$ be any function which is unbounded above, e.g.

$$f(x) := \begin{cases} \frac{1}{x} & \text{for } 0 < x \leq 1; \\ 0 & \text{for } x = 0. \end{cases}$$

For any $n \in \mathbb{N}$, define $f_n : [0,1] \to \mathbb{R}$ by $f_n(x) := \min(f(x), n)$. Then f_n is continuous except at $x = 0$, hence integrable. But $f_n \to f$ pointwise, since for any $0 \leq x \leq 1$ we have $f_n(x) = f(x)$ for $n > f(x)$, and f is not integrable because it is unbounded.

It is also possible to give an example where f is bounded. Choose a bijection $\mathbb{N} \xrightarrow{\sim} \mathbb{Q} \cap [0,1]$, which is possible since $\mathbb{Q} \cap [0,1]$ is countably infinite. This amounts to a sequence which contains every element of $\mathbb{Q} \cap [0,1]$ exactly once:

$$\mathbb{Q} \cap [0,1] = \{a_1, a_2, \cdots\}.$$

For any $n \in \mathbb{N}$, define $f_n : [0,1] \to \mathbb{R}$ by

$$f_n(x) := \begin{cases} 1 & \text{if } x = a_k \text{ for some } 1 \le k \le n, \\ 0 & \text{otherwise.} \end{cases}$$

Each f_n is integrable with integral 0, since it differs from the zero function at only finitely many points. But we have $f_n \to f$ pointwise where $f : [0,1] \to \mathbb{R}$ is defined by

$$f(x) := \begin{cases} 1 & \text{if } x \in \mathbb{Q}, \\ 0 & \text{if } x \notin \mathbb{Q}, \end{cases}$$

because for any $x \in \mathbb{Q} \cap [0,1]$ there exists $N \in \mathbb{N}$ such that $x = a_N$, and hence $n > N$ implies that $x \in \{a_1, \cdots, a_n\}$. We saw in class that f is not integrable.

2. (a) Apply the ratio test:

$$\lim_{k\to\infty} \frac{\left|\binom{\alpha}{k+1} x^{k+1}\right|}{\left|\binom{\alpha}{k} x^k\right|} = |x| \lim_{k\to\infty} \left|\frac{\alpha - k}{k+1}\right| = |x|.$$

So the power series is absolutely convergent, hence convergent, for $|x| < 1$.

(b) Recall that we can differentiate a power series term-by-term in an open interval where it converges. Thus we can compute

$$\begin{aligned}(1+x)f'(x) &= (1+x)\sum_{k=1}^{\infty} \binom{\alpha}{k} k x^{k-1} \\ &= \sum_{k=1}^{\infty} \binom{\alpha}{k} k x^{k-1} + \sum_{k=1}^{\infty} \binom{\alpha}{k} k x^k \\ &= \alpha + \sum_{k=1}^{\infty} \left(\binom{\alpha}{k+1}(k+1) + \binom{\alpha}{k} k \right) x^k.\end{aligned}$$

Now observe that for any $k \in \mathbb{N}$, we have

$$\begin{aligned}\binom{\alpha}{k+1}(k+1) + \binom{\alpha}{k} k &= \frac{\alpha \cdots (\alpha - k)}{k!} + \frac{\alpha \cdots (\alpha - k + 1)}{(k-1)!} \\ &= \alpha \left(\frac{(\alpha-1)\cdots(\alpha-k)}{k!} + \frac{(\alpha-1)\cdots(\alpha-k+1)}{(k-1)!} \right) \\ &= \alpha \left(\binom{\alpha-1}{k} + \binom{\alpha-1}{k-1} \right) \\ &= \alpha \binom{\alpha}{k}.\end{aligned}$$

In the last step, we used a standard identity which was established in Problem Set 2 from 161 (there α was a nonnegative integer, but the same calculation proves the identity in general). Continuing the calculation from before, we obtain

$$\begin{aligned}(1+x)f'(x) &= \alpha + \sum_{k=1}^{\infty}\left(\binom{\alpha}{k+1}(k+1) + \binom{\alpha}{k}k\right)x^k \\ &= \alpha + \alpha\sum_{k=1}^{\infty}\binom{\alpha}{k}x^k \\ &= \alpha\sum_{k=0}^{\infty}\binom{\alpha}{k}x^k = \alpha f(x).\end{aligned}$$

(c) As suggested, we consider the function $g : (-1,1) \to \mathbb{R}$ defined by $g(x) := f(x)/(1+x)^\alpha$. By the power rule, which we proved for arbitrary exponents, we have

$$\frac{\mathrm{d}}{\mathrm{d}x}(1+x)^\alpha = \alpha(1+x)^{\alpha-1}.$$

Now apply the quotient rule to see that

$$g'(x) = \frac{f'(x)(1+x)^\alpha - f(x)\alpha(1+x)^{\alpha-1}}{(1+x)^{2\alpha}} = \frac{\alpha f(x)(1+x)^{\alpha-1} - f(x)\alpha(1+x)^{\alpha-1}}{(1+x)^{2\alpha}} = 0,$$

where we applied part (b) for the second equality. It follows that $g(x) = C$, i.e. $f(x) = C(1+x)^\alpha$, for some $C \in \mathbb{R}$. Substitute $x = 0$ to obtain

$$1 = f(0) = C(1+0)^\alpha = C$$

and hence $f(x) = (1+x)^\alpha$.

3. (a) Since $a \neq 0$, we can write $f(x) = -\frac{1}{a}\frac{1}{1-x/a}$. Then the formula for geometric series yields

$$f(x) = -\frac{1}{a}\left(1 + \frac{x}{a} + \frac{x^2}{a^2} + \cdots\right) = -\frac{1}{a} - \frac{x}{a^2} - \frac{x^2}{a^3} - \cdots$$

for $|\frac{x}{a}| < 1$, i.e. $|x| < |a|$.

(b) We have $f'(x) = \frac{1}{x-a}$ and hence

$$\int_0^x \frac{\mathrm{d}t}{t-a} = \log(t-a)\Big|_0^x = \log(x-a) - \log(-a).$$

Since the power series in (a) converges uniformly on $[-r, r]$ for any $0 < r < |a|$, we can integrate it term by term to obtain

$$\begin{aligned}\log(x-a) - \log(-a) &= \int_0^x \frac{\mathrm{d}t}{t-a} \\ &= \int_0^x \left(-\frac{1}{a} - \frac{t}{a^2} - \frac{t^2}{a^3} - \cdots\right)\mathrm{d}t \\ &= -\frac{x}{a} - \frac{x^2}{2a^2} - \frac{x^3}{3a^3} - \cdots .\end{aligned}$$

Thus the desired Taylor series is

$$\log(x-a) = \log(-a) - \frac{x}{a} - \frac{x^2}{2a^2} - \frac{x^3}{3a^3} - \cdots,$$

valid for $|x| < -a$.

(c) Apply #2 to obtain

$$f(x) = (1-x)^{-1/2} = \sum_{k=0}^{\infty} \binom{-1/2}{k} x^k,$$

valid for $|x| < 1$. To give a feeling for how these binomial coefficients look, we point out that

$$\binom{-1/2}{k} = \frac{(-1/2)(-3/2)\cdots(1-2k)/2}{k!} = (-1)^k \frac{1 \cdot 3 \cdots (2k-1)}{k! \cdot 2^k}.$$

(d) Substitute x^2 into the power series from part (c), which yields

$$f(x) = \sum_{k=0}^{\infty} \binom{-1/2}{k} x^{2k}.$$

This Taylor series converges for $x^2 < 1$, which is equivalent to $|x| < 1$.

(e) Analogously to part (b), we can integrate the power series from (d) term by term to obtain

$$\arcsin x = \int_0^x \frac{\mathrm{d}t}{\sqrt{1-t^2}} = \int_0^x \sum_{k=0}^{\infty} \binom{-1/2}{k} t^{2k}\, \mathrm{d}t = \sum_{k=0}^{\infty} \binom{-1/2}{k} \frac{x^{2k+1}}{2k+1},$$

again valid for $|x| < 1$.

4. (a) Clearly (a_n) is an increasing sequence, i.e. $a_{n-1} \le a_n$ for all $n > 1$. Thus we have

$$\frac{a_{n+1}}{a_n} = \frac{a_n + a_{n-1}}{a_n} = 1 + \frac{a_{n-1}}{a_n} \le 2.$$

(b) Since $a_{n+1} \le 2a_n$ by part (a) and $a_1 = 1$, a simple induction shows that $a_n \le 2^n$ for any $n \in \mathbb{N}$. Thus we have

$$|a_n x^{n-1}| = |a_n||x|^{n-1} \le 2^n |x|^{n-1}.$$

The geometric series

$$\sum_{n=1}^{\infty} 2^n |x|^{n-1} = 2 \sum_{n=0}^{\infty} (2|x|)^n$$

converges for $2|x| < 1$, i.e. $|x| < \frac{1}{2}$. Thus the comparison test shows that the series $f(x)$ converges (absolutely) for $|x| < \frac{1}{2}$.

(c) Observe that

$$\begin{aligned} xf(x) + x^2 f(x) &= \sum_{n=1}^{\infty} a_n x^n + \sum_{n=1}^{\infty} a_n x^{n+1} \\ &= a_1 x + \sum_{n=2}^{\infty} (a_n + a_{n-1}) x^n \\ &= a_2 x + \sum_{n=2}^{\infty} a_{n+1} x^n \\ &= \sum_{n=2}^{\infty} a_n x^{n-1} = f(x) - 1. \end{aligned}$$

Solving for $f(x)$, we obtain

$$f(x) = \frac{-1}{x^2 + x - 1}$$

for $|x| < \frac{1}{2}$ as desired (dividing by $x^2 + x - 1$ is valid because its roots $\frac{-1 \pm \sqrt{5}}{2}$ lie outside of the interval $(-\frac{1}{2}, \frac{1}{2})$).

(d) First we find the partial fraction decomposition for $\frac{1}{x^2+x-1}$. Write $\varphi := \frac{1+\sqrt{5}}{2}$, $\psi := \frac{1-\sqrt{5}}{2}$, so that

$$x^2 + x - 1 = (x+\varphi)(x+\psi).$$

We must solve the equation

$$\frac{1}{x^2+x-1} = \frac{A}{x+\varphi} + \frac{B}{x+\psi}$$

for A and B. Clearing denominators and rearranging, this equation becomes

$$(A+B)x + A\psi + B\varphi = 1,$$

which is equivalent to the system of equations

$$\begin{aligned} A + B &= 0, \\ A\psi + B\varphi &= 1. \end{aligned}$$

Solve this system to obtain $A = -\frac{1}{\sqrt{5}}$ and $B = \frac{1}{\sqrt{5}}$. Thus we have

$$\frac{1}{x^2+x-1} = \frac{1}{\sqrt{5}}\left(-\frac{1}{x+\varphi} + \frac{1}{x+\psi}\right),$$

or equivalently

$$f(x) = \frac{-1}{x^2+x-1} = \frac{1}{\sqrt{5}}\left(\frac{1}{x+\varphi} - \frac{1}{x+\psi}\right).$$

Next, we apply #3(a) to obtain the power series expansion

$$\frac{1}{x+\varphi} = \frac{1}{\varphi} + \frac{x}{\varphi^2} + \frac{x^2}{\varphi^3} + \cdots,$$

valid for $|x| < \varphi$. Note that $\varphi\psi = -1$, so we can rewrite this as

$$\frac{1}{x+\varphi} = -\psi - \psi^2 x - \psi^3 x^2 - \cdots.$$

Analogously, we find that

$$\frac{1}{x+\psi} = -\varphi - \varphi^2 x - \varphi^3 x^2 - \cdots,$$

valid for $|x| < |\psi| = \frac{1}{\varphi}$. Combining this with the above partial fraction decomposition yields

$$\begin{aligned} f(x) &= \frac{1}{\sqrt{5}}\left(\frac{1}{x+\varphi} - \frac{1}{x+\psi}\right) \\ &= \frac{1}{\sqrt{5}}(-\psi - \psi^2 x - \psi^3 x^2 - \cdots - (-\varphi - \varphi^2 x - \varphi^3 x^2 - \cdots)) \\ &= \frac{1}{\sqrt{5}}(\varphi - \psi + (\varphi^2 - \psi^2)x + (\varphi^3 - \psi^3)x^2 + \cdots), \end{aligned}$$

valid for $|x| < \min(\varphi, \frac{1}{\varphi}) = \frac{1}{\varphi}$.

(e) Both power series expressions for $f(x)$ are valid for $|x| < \min(\frac{1}{2}, \frac{1}{\varphi}) = \frac{1}{2}$, and hence their coefficients must agree by the uniqueness statement proved in class. The coefficient of x^{n-1} in the power series expansion for $f(x)$ obtained in part (d) is $\frac{1}{\sqrt{5}}(\varphi^n - \psi^n)$, which must therefore equal a_n as claimed.

5. (a) For the first identity, apply integration by parts and then the Pythagorean theorem to obtain

$$\begin{aligned}
\int_0^{\frac{\pi}{2}} \cos^n t \,\mathrm{d}t &= \cos^{n-1} t \sin t \Big|_0^{\frac{\pi}{2}} + (n-1)\int_0^{\frac{\pi}{2}} \cos^{n-2} t \sin^2 t \,\mathrm{d}t \\
&= (n-1)\int_0^{\frac{\pi}{2}} \cos^{n-2} t \sin^2 t \,\mathrm{d}t \\
&= (n-1)\int_0^{\frac{\pi}{2}} \cos^{n-2} t \,\mathrm{d}t - (n-1)\int_0^{\frac{\pi}{2}} \cos^{n} t \,\mathrm{d}t\,.
\end{aligned}$$

Add $(n-1)\int_0^{\frac{\pi}{2}} \cos^n t \,\mathrm{d}t$ to both sides and then divide by n to obtain the desired identity. Now a simple induction shows that

$$\begin{aligned}
\int_0^{\frac{\pi}{2}} \cos^{2n} t \,\mathrm{d}t &= \frac{2n-1}{2n}\int_0^{\frac{\pi}{2}} \cos^{2n-2} t \,\mathrm{d}t \\
&= \frac{(2n-1)(2n-3)}{2n(2n-2)}\int_0^{\frac{\pi}{2}} \cos^{2n-4} t \,\mathrm{d}t \\
&= \cdots = \prod_{k=1}^{n} \frac{2k-1}{2k}\int_0^{\frac{\pi}{2}} \mathrm{d}t \\
&= \frac{\pi}{2}\prod_{k=1}^{n} \frac{2k-1}{2k}.
\end{aligned}$$

Similarly, we have

$$\begin{aligned}
\int_0^{\frac{\pi}{2}} \cos^{2n+1} t \,\mathrm{d}t &= \frac{2n}{2n+1}\int_0^{\frac{\pi}{2}} \cos^{2n-1} t \,\mathrm{d}t \\
&= \frac{2n(2n-2)}{(2n+1)(2n-1)}\int_0^{\frac{\pi}{2}} \cos^{2n-3} t \,\mathrm{d}t \\
&= \cdots = \prod_{k=1}^{n} \frac{2k}{2k+1}\int_0^{\frac{\pi}{2}} \cos t \,\mathrm{d}t \\
&= \prod_{k=1}^{n} \frac{2k}{2k+1} \sin t \Big|_0^{\frac{\pi}{2}} \\
&= \prod_{k=1}^{n} \frac{2k}{2k+1}.
\end{aligned}$$

(b) Since $0 \le \cos t \le 1$ for $0 \le t \le \frac{\pi}{2}$, we have

$$0 \le \cos^{2n+1} t \le \cos^{2n} t$$

and hence

$$0 \le \int_0^{\frac{\pi}{2}} \cos^{2n+1} t \,\mathrm{d}t \le \int_0^{\frac{\pi}{2}} \cos^{2n} t \,\mathrm{d}t$$

for any $n \in \mathbb{N}$. Moreover, since $\cos^{2n} 0 = 1$ and $\cos^{2n} t$ is continuous, we have $\int_0^{\frac{\pi}{2}} \cos^{2n} t \,\mathrm{d}t > 0$. Thus we can divide the previous inequality by this integral to obtain

$$\frac{\int_0^{\frac{\pi}{2}} \cos^{2n+1} t \,\mathrm{d}t}{\int_0^{\frac{\pi}{2}} \cos^{2n} t \,\mathrm{d}t} \le 1.$$

For the other inequality, we observe that

$$\frac{2n+1}{2n+2}\int_0^{\frac{\pi}{2}} \cos^{2n} t \,\mathrm{d}t = \int_0^{\frac{\pi}{2}} \cos^{2n+2} t \,\mathrm{d}t \le \int_0^{\frac{\pi}{2}} \cos^{2n+1} t \,\mathrm{d}t$$

by the reduction formula from (a), so that

$$\frac{2n+1}{2n+2} \leq \frac{\int_0^{\frac{\pi}{2}} \cos^{2n+1} t \,\mathrm{d}t}{\int_0^{\frac{\pi}{2}} \cos^{2n} t \,\mathrm{d}t}.$$

(c) Using the product formulas from (a), we obtain

$$\frac{\int_0^{\frac{\pi}{2}} \cos^{2n+1} t \,\mathrm{d}t}{\int_0^{\frac{\pi}{2}} \cos^{2n} t \,\mathrm{d}t} = \frac{2}{\pi} \prod_{k=1}^{n} \frac{(2k)^2}{(2k+1)(2k-1)}.$$

Now the inequalities from (b) yield

$$\frac{2n+1}{2n+2} \leq \frac{2}{\pi} \prod_{k=1}^{n} \frac{(2k)^2}{(2k+1)(2k-1)} \leq 1.$$

Passing to the limit as $n \to \infty$, we have

$$\lim_{n\to\infty} \frac{2n+1}{2n+2} = 1$$

and hence

$$\frac{2}{\pi} \prod_{k=1}^{\infty} \frac{(2k)^2}{(2k+1)(2k-1)} = 1$$

by the squeeze theorem. Multiply by $\frac{\pi}{2}$ to finish the proof.